Stability of immune globulin intravenous, human-stwk, 10% liquid following pooling and storage in flexible containers

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ABSTRACT

Background: Immune globulin intravenous, human-stwk, 10% liquid is indicated for the treatment of primary immunodeficiency disorders associated with defects in humoral immunity. Unopened immune globulin intravenous, human-stwk, 10% liquid in its original container is to be stored under refrigerated conditions at (2-8°C/36-46°F) for 36 months or may be stored for 24 months at room temperature (25°C/77°F) from the date it was manufactured. However, pooling intravenous immunoglobulin (IVIG) offers streamlining of administration for the administering nurse, improves the patient's ability to ambulate freely during the infusion, and a provides reduced risk of container breakage during shipment and administration. Therefore, it is common practice to pool vials of IVIG into a single container (usually a flexible bag) under aseptic conditions. Although the United States Pharmacopeia (USP), a scientific nonprofit organization, regulates the compounding of sterile products via the USP <797> Compounding Standards, it is important to understand individual product stability under these common conditions.

Purpose: The purpose of this study was to determine the effects of pooling immune globulin intravenous, human-stwk, 10% liquid in three types of plastic containers under aseptic conditions.

Methods: Pooling of the immune globulin intravenous, human-stwk, 10% liquid was performed aseptically under a laminar flow hood. The product was pooled into an Empty IntraVia Container (Baxter), an EVA 2-port Container (The Metrix Company), and a 3-in-1 EVA Mixing Container (B. Braun). These pooled samples were compared to samples obtained from product in the original manufacturer containers, and whenever possible, the original release specifications. The following attributes were evaluated to determine product stability and quality at days 0, 1, 7, and 14: pH, total protein composition, osmolality, molecular size distribution (IgG monomers plus dimers, IgG fragments, IgG polymers), anti-complement activity, hepatitis B antibody titer, appearance, particle matter, and TGA. After completion of the study, sterility testing was performed to verify that aseptic technique was maintained throughout the entire study.

Results: Pooled immune globulin intravenous, human-stwk, 10% liquid remained within release specifications (which are the FDA-Approved requirements for immune globulin intravenous, human-stwk, 10% liquid) for all parameters evaluated for 14 days when stored under refrigerated conditions (2°-8°C/36°-46°F).

Discussion and conclusion: Pooled IVIG products fall into USP <797> Category 2, which indicates that when product pooling takes place in a modern clean room equipped with an ISO Class 5 Primary Engineering Control (PEC), an ISO Class 7 buffer area, and an ISO Class 8 ante-room, the assigned beyond use date for that product is 4 days at controlled room temperature and 10 days when refrigerated. Nonetheless, it is important to understand individual product characteristics under these conditions. In this in vitro study, immune globulin intravenous, human-stwk, 10% liquid remained within FDA approved release specifications for all parameters evaluated for 14 days when stored under refrigerated conditions (2°-8°C/36°-46°F).

BACKGROUND

Administration of intravenous immune globulin (IVIG) commonly involves pooling single-dose vials into a single container (usually a flexible bag) under aseptic conditions. This practice has several advantages:

- ✓ It streamlines the administration process for the administering nurse
- ✓ It improves the patient's ability to ambulate freely during infusion
- ✓ It reduces the risk of container breakage during shipment and administration

Immune globulin intravenous, human-stwk, 10% liquid is indicated for the treatment of adults with primary humoral immunodeficiency and is supplied in single-dose, tamper-evident vials.[1] Unopened vials may be stored under refrigerated conditions (2-8°C/36-46°F) for 36 months or for 24 months at room temperature (25°C/77°F) from the date of manufacture.[1]

ABBREVIATIONS

Anti-HBs: Hepatitis B antibody titer
IgG: Immune globulin G
IVIG: Intravenous immune globulin
TGA: Thrombin generation assay
USP: US Pharmacopeia

PURPOSE

In this study, we sought to assess the stability of this product after pooling for up to 14 days to confirm safety and potency under conditions reflecting clinical practice.

METHODS

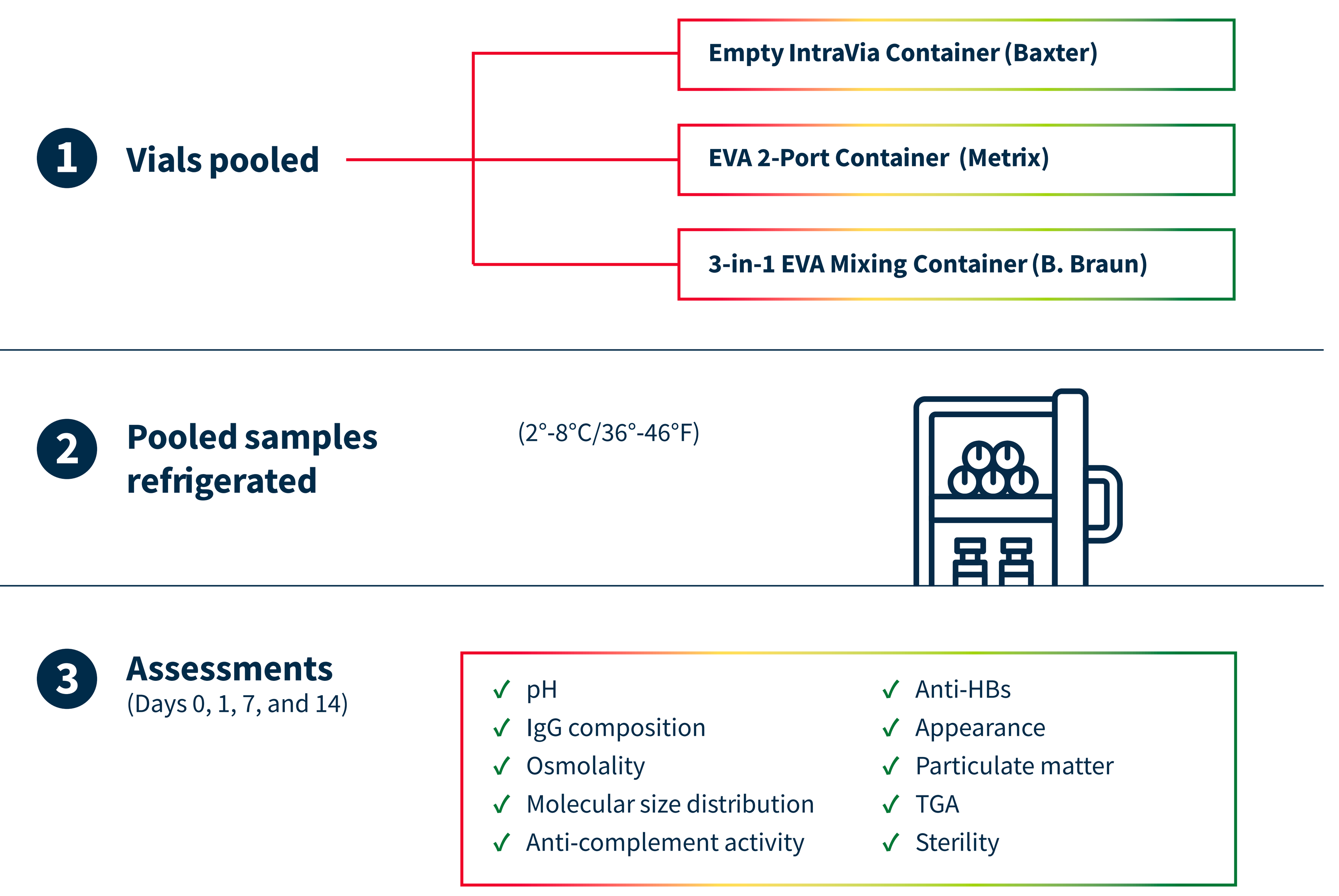
Pooling of ALYGLO® immune globulin intravenous, human-stwk, 10% liquid was performed aseptically under a laminar flow hood into 3 different containers:

1. Empty IntraVia Container (Baxter)
2. EVA 2-Port Container (Metrix)
3. 3-in-1 EVA Mixing Container (B. Braun) (Figure 1)

The pooling volume was determined based on the assumption that a 70 kg adult receiving a 300 mg/kg dose would require 21 g, or 210 mL, of immune globulin intravenous, human-stwk, 10% liquid. 2 x 200 mL vials and 3 x 100 mL vials were pooled separately for each different container. Samples were collected from containers using an infusion kit.

Pooled samples were compared to product from the original manufacturer containers. The following attributes were evaluated to determine product stability and quality at Days 0, 1, 7, and 14: pH, IgG composition, osmolality, molecular size distribution, anti-complement activity, hepatitis B antibody titer (anti-HBs), appearance, particulate matter, and thrombin generation assay (TGA). Sterility testing was performed to verify that aseptic technique was maintained throughout the entire study.

FIGURE 1.


1
Vials pooled
Empty IntraVia Container (Baxter)
EVA 2-Port Container (Metrix)
3-in-1 EVA Mixing Container (B. Braun)
2
Pooled samples refrigerated
(2°-8°C/36°-46°F)
3
Assessments
(Days 0, 1, 7, and 14)
✓ pH
✓ IgG composition
✓ Osmolality
✓ Molecular size distribution
✓ Anti-complement activity
✓ Anti-HBs
✓ Appearance
✓ Particulate matter
✓ TGA
✓ Sterility

RESULTS

- ✓ Representative results from samples drawn from immune globulin intravenous, human-stwk, 10% liquid pooled in the Empty IntraVia Container (Baxter) are shown in Table 1. Product release criteria are cited in the first column.
- ✓ **No assessment from any pooled sample from any of the 3 containers differed from product release specifications at any time point.**
- ✓ Sterility was confirmed at each time point for both pooled samples for all 3 containers.

TABLE 1. Test results for the 200 mL vials pooled in the Empty IntraVia Container (Baxter)

Assessment	Release criteria	Day 0	Day 1	Day 7	Day 14
Appearance (liquid)	Colorless or pale yellow, clear or slightly opalescent	Pale yellow, slightly opalescent	Pale yellow, slightly opalescent	Pale yellow, slightly opalescent	Pale yellow, slightly opalescent
pH	4.5-5.5	5.03	5.04	5.03	5.02
IgG composition	≥96% of total protein	100	100	100	100
Osmolality	240-360 mOsmol/kg	289	291	290	289
Molecular size distribution	Monomer + dimer ≥92%	99.0	98.93	98.86	98.83
	Polymers ≤3%	<0.3	<0.3	<0.3	<0.3
	Fragments ≤5%	0.96	1.03	1.10	1.12
Anti-complement activity	Complement consumption ≤50%	26.96	25.75	28.12	23.88
Anti-HBs	≥1.0 IU/g IgG	61.51	60.50	62.54	62.59
Particulate matter	No visible particles	None	None	None	None
TGA	≤3.12 mU/mL	<1.56*	<1.56	<1.56	<1.56
Sterility	Sterile	Sterile	N/T†	Sterile	Sterile

*TGA assay detection limit; †Not tested

DISCUSSION

The US Pharmacopeia (USP) sets drug quality standards recognized by the FDA as the definitive source for drug safety requirements. The USP standards for aseptically pooled IVIG products assign a "beyond use" date of 4 days at controlled room temperature, and 10 days when refrigerated.[3] We sought to assess the stability of immune globulin intravenous, human-stwk, 10% liquid after 14 days under refrigerated conditions. In this rigorous study, immune globulin intravenous, human-stwk, 10% liquid remained within FDA-approved release specifications for all parameters evaluated for 14 days when stored under refrigerated conditions in 3 different commercially available containers.

REFERENCES

1. ALYGLO Prescribing Information. GC Biopharma; 2023. **2.** Immunoglobulin National Society. *Immunoglobulin Therapy Standards of Practice.* 3.2 ed. IgNS; 2024. Accessed March 11, 2025. https://ig-ns.org/product/ig-therapy-standards-of-practice-version-3-2 **3.** USP Compounding Compendium. USP. Accessed March 11, 2025. https://www.usp.org/products/usp-compounding-compendium **4.** Data on file. GC Biopharma; 2024.